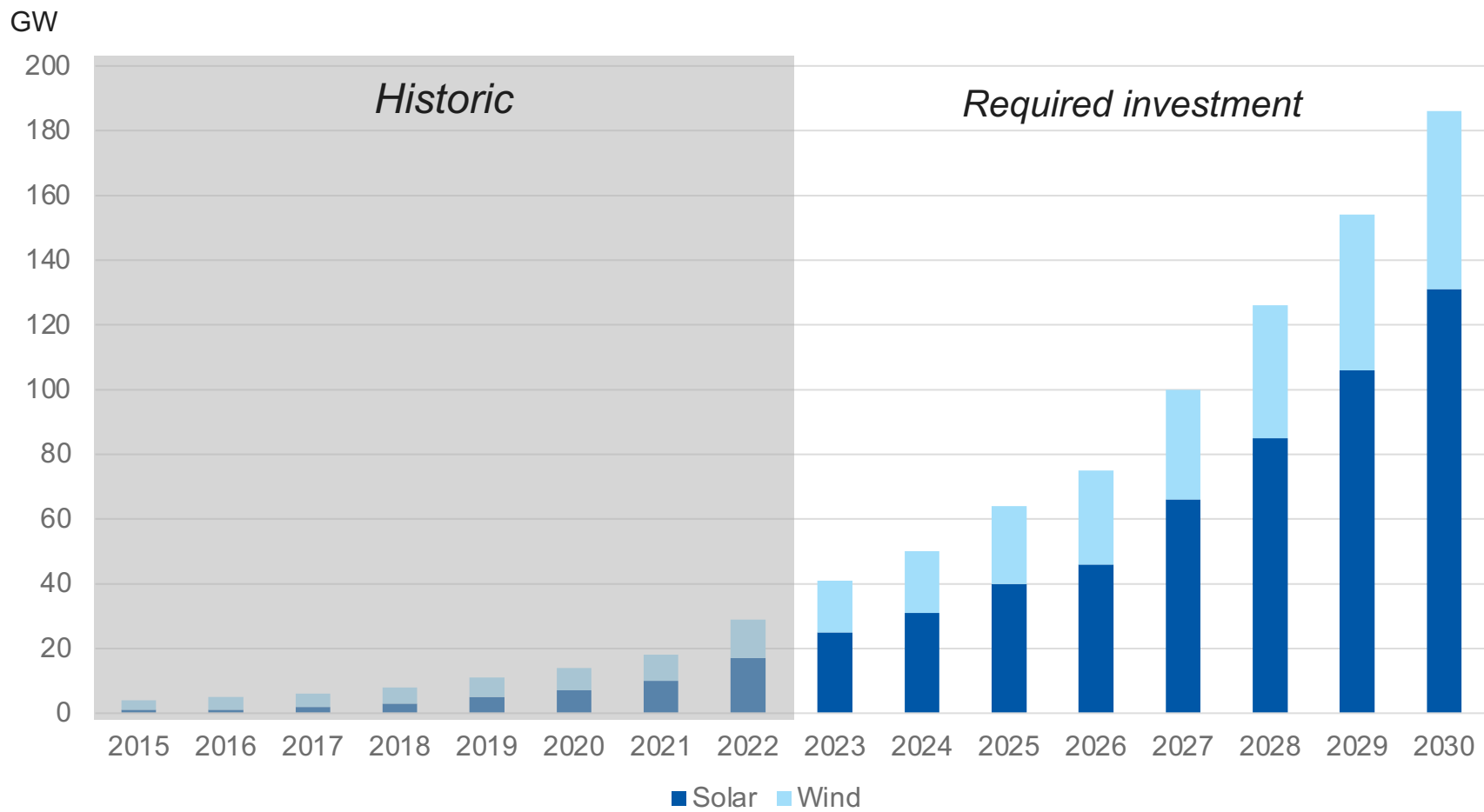


Renewable Backed Securities:
**Accelerating Africa's
Energy Transition**



Rapid growth of utility-scale RES projects in Africa is required to meet climate goals

Solar and Wind Capacity in Africa, *IEA Africa energy outlook 2022*

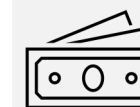


Many African initiatives focus on small scale, residential and off-grid solutions

While helpful, these solutions **cannot provide the required scale** for a renewable based economy



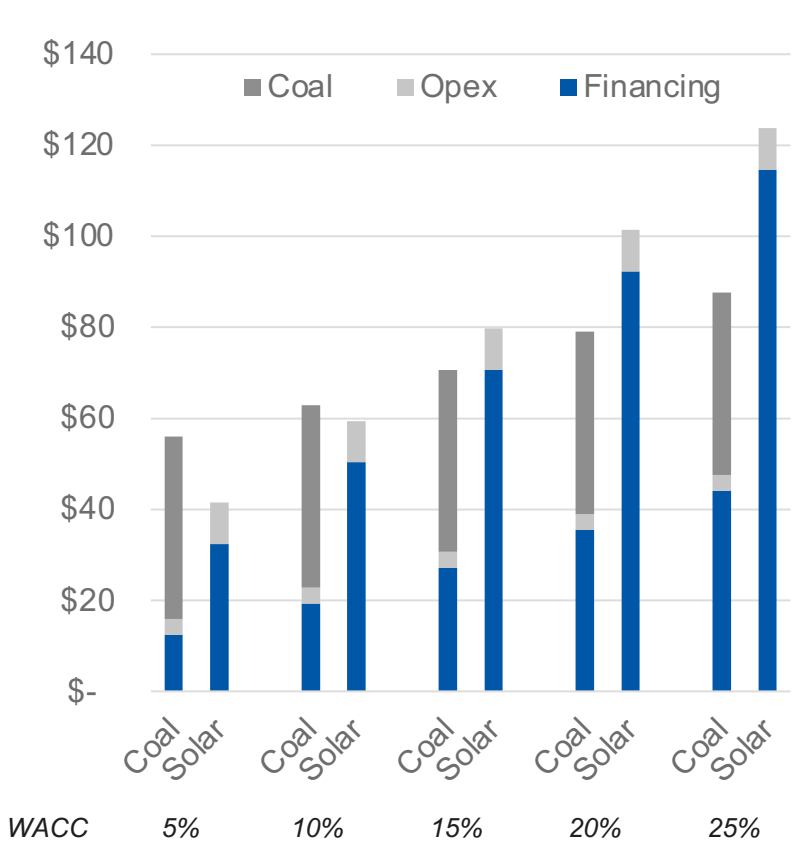
Only **Utility scale projects can ensure** low-cost energy for future generations and provide long-term sustainable economic growth



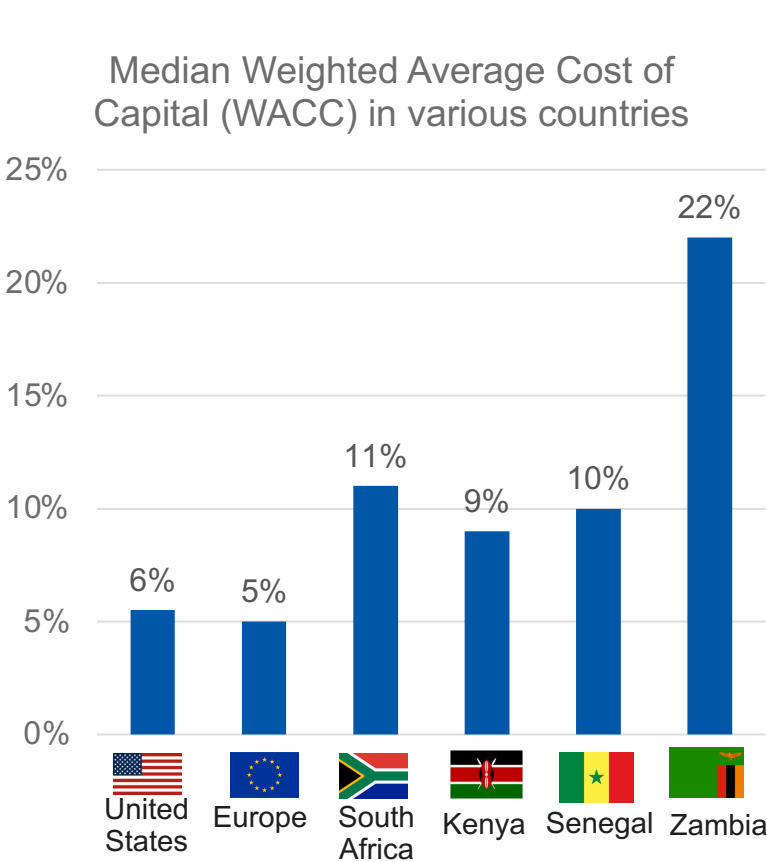
These projects will require massive private **investments of ~130 USD bln** annually from 2026 onward

Cost of Capital determines RES competitiveness... ...and it is too high in Africa

LCOE breakdown for different cost of capital, USD/MWh



Cost of Capital² guaranteed cashflow infra-projects in selected countries, IEA 2022 & 2023, Climate Policy Initiative



Higher cost of capital renders fossil projects **more viable** in all **African countries** than renewable projects



In “more investible” African countries, viability **hinges on percentage points differences**

Higher cost of capital is driven by **currency, sovereign and political risk**



These risks persist in the operational phase, making **operational viability the bottleneck** of renewables in Africa

1.Levelized Cost of Electricity, US\$/MWh 2. Construction phase

We believe part of the solution lies in securitizing renewable projects across Africa

Securitization lowers cost of capital through two distinct and indirect market mechanisms...

... it further provides needed velocity to investment and enables more effective multilateral funding

1 Risk reduction "pass-down" effects:



2 Increased capital supply effect:



3 Increased velocity of lending:

- Africa requires \$130 bln p.a. in clean energy investment from 2026 onward
- A mature securitization market allows (development) banks to offload loans from their balance sheet
- Enabling banks to reinvest capital and improve the velocity of lending to renewable power projects

4 Improved effectiveness of multilateral funding:

- Increased private capital supply lowers the share of public capital needed for an individual product
- Leading to a larger realized capacity and impact per (multilateral) invested dollar
- Based on a 270 million participation, a back of the envelope calculation translates into a 10x magnitude impact



Our product would **expand the adoption of clean energy**



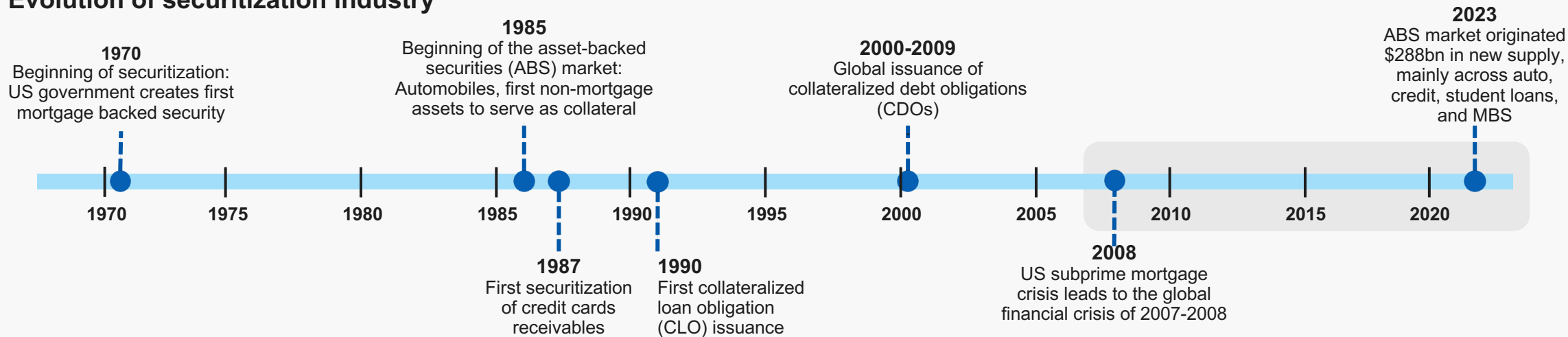
As RBS goes into its subsequent iterations, the result would be a **decrease in economies' carbon intensity**



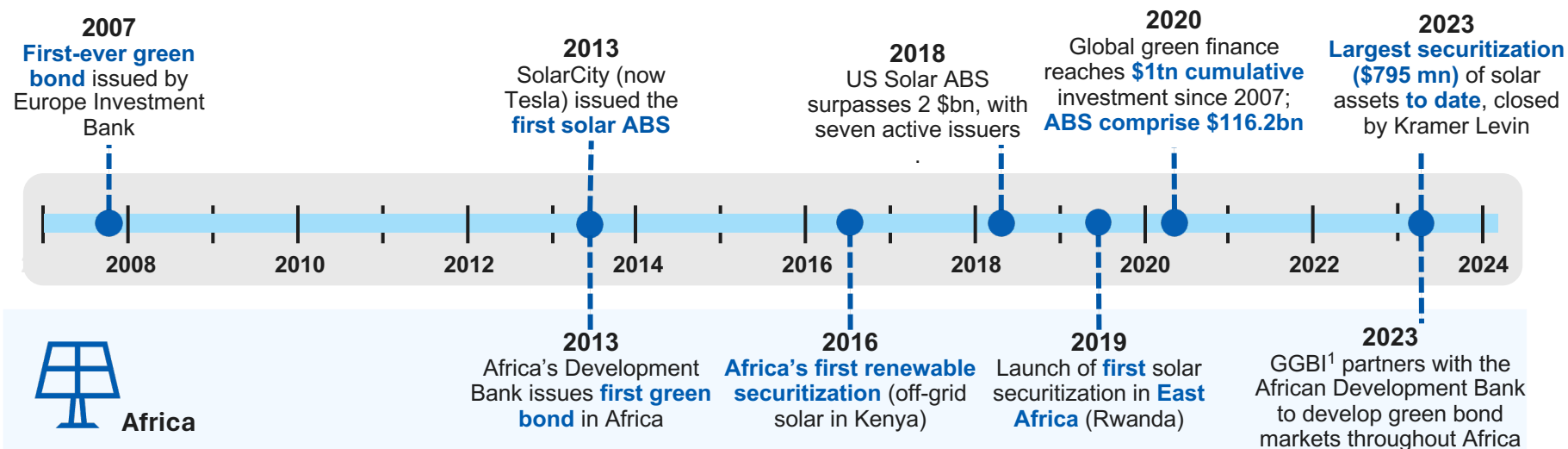
It is necessary for the success of our venture to bring in **partners along each step of our value chain**

Securitization is a proven market solution that is evolving rapidly to enhance the flow and cost of capital for sustainable projects

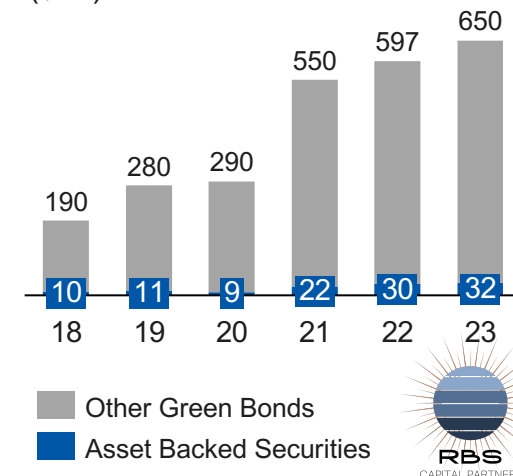
Evolution of securitization industry



Evolution of green securitization



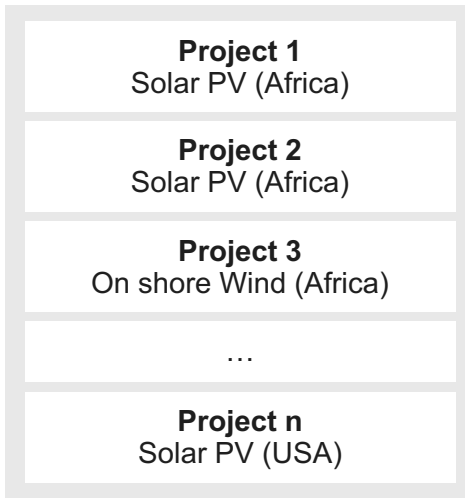
Global green bond investment (\$bn)



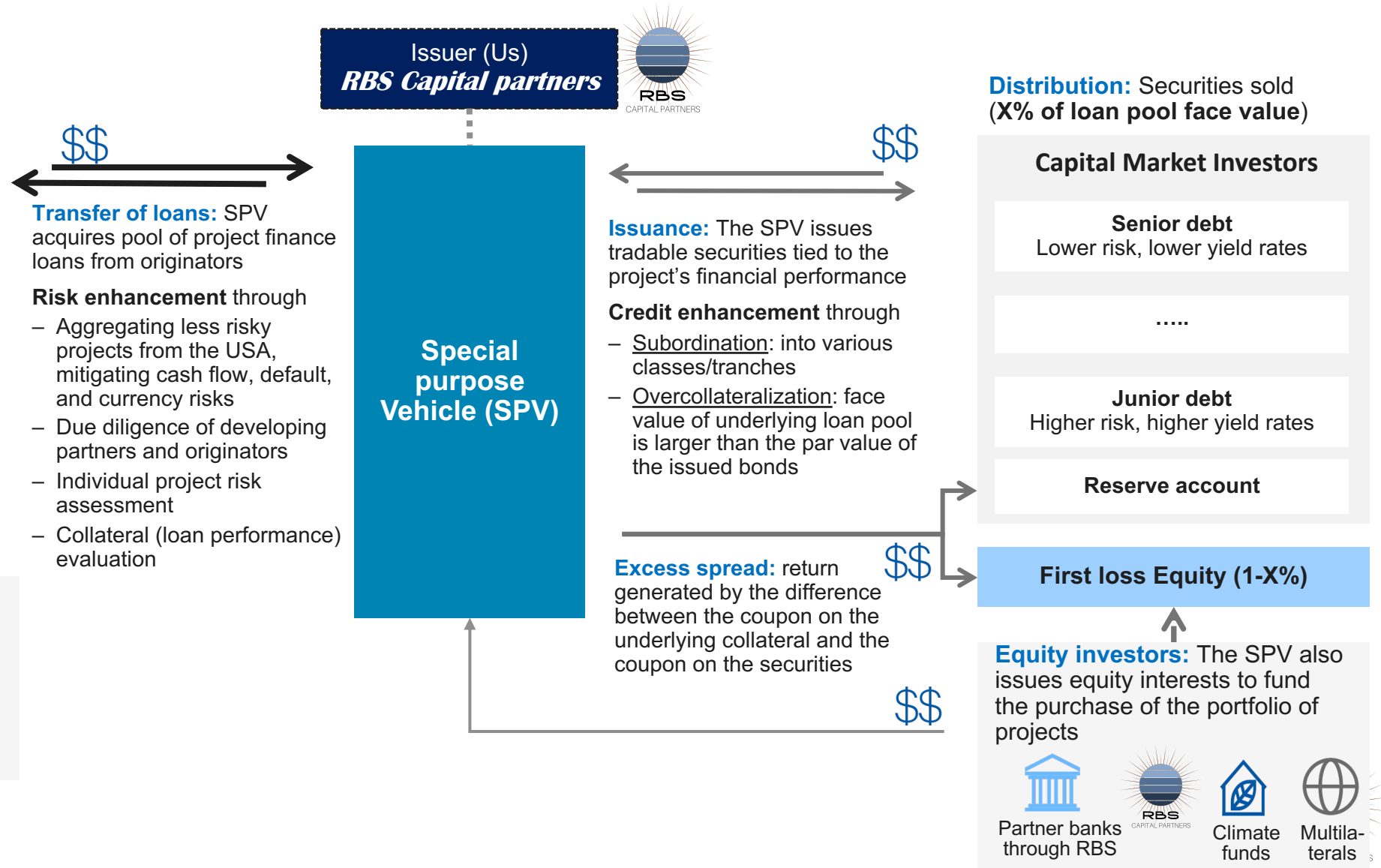
RBS Capital: pooling RES debt to create ABS that provide attractive risk-adjusted returns & accelerate Africa's energy transition

RBS at a glance

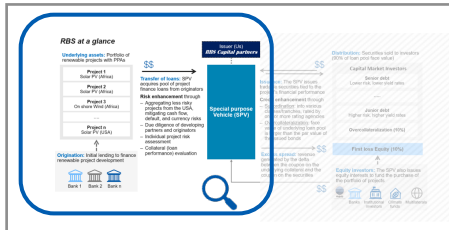
Underlying assets: Portfolio of renewable projects with PPAs



Origination: Initial lending to finance renewable project development



Our fund underwrites \$3bn of selected operational projects in Africa and the USA



Underlying assets: portfolio of 20 renewable projects

Solar 100MW - Africa

70% debt financed
Principal – \$84mn

Wind 140MW - Africa

80% debt financed
Principal – \$233mn

Solar 200MW - Africa

76% debt financed
Principal – \$180mn

....

Solar 250MW - USA

70% debt financed
Principal – \$140mn

Transfer of loans: SPV acquires pool of project finance loans from originators

- Principal of **USD 3 billion**
- Loan interest @ **SOFR + 6.0%**¹
 - Africa: @ 14.0%
 - USA: @ SOFR + 2.0%

Origination: Initial lending to finance renewable project development

- **60%** debt from Africa
- **40%** debt from USA



Issuer (Us)
RBS Capital partners

Special purpose Vehicle (SPV)

Selected projects originators



Standard Bank
(South Africa)



Nedbank
(South Africa)



Absa Bank
(Kenya)



International
Finance
Corporation



US Lenders

Considerations

- **Large size fund + utility scale focus**
 - SPV success requires targeting a **diversified portfolio of at least 20 projects**. Utility projects scale (avg. \$150mn debt size) demands such large investment
 - Utility scale projects present **less risk compared to off-grid/residential** due to more stable revenues, off-take agreements, and clearer regulation. Cash flow collection from off-takers directly
- **Combined USA and African projects**
 - **Risk mitigation:** The US provides a platform to aggregate less risky projects, mitigating cash flow, default, and currency risks from African projects
 - **Attractiveness:** this blend appeals to a wider range of investors, looking for high-impact opportunities and the security and predictability of the US market
- **Rigorous project selection**
 - **Exhaustive vetting process:** size of portfolio (20 projects) allows for dedicated due diligence process
 - Geographical criteria: Low macro and political risks, pegged or dollar denominated currencies
 - Individual project risk assessment (inc. lender, sponsor, off-take agreements; multilateral and export credit enhancements)
 - **Operational status:** further risk mitigation targeting projects generating immediate cashflows
- **Transfer of loans**
 - **Partnerships** with trusted developers and lenders mitigates adverse selection/asymmetry of information; initial list of lenders identified
 - **Target loan profile:** acquisition of most senior loan
 - Africa: 11-15% all-in interest rate
 - US: 3-4% spread over SOFR
 - Tenor: 15-20 years
 - **Early buy-back penalty:** 1.5% of principal



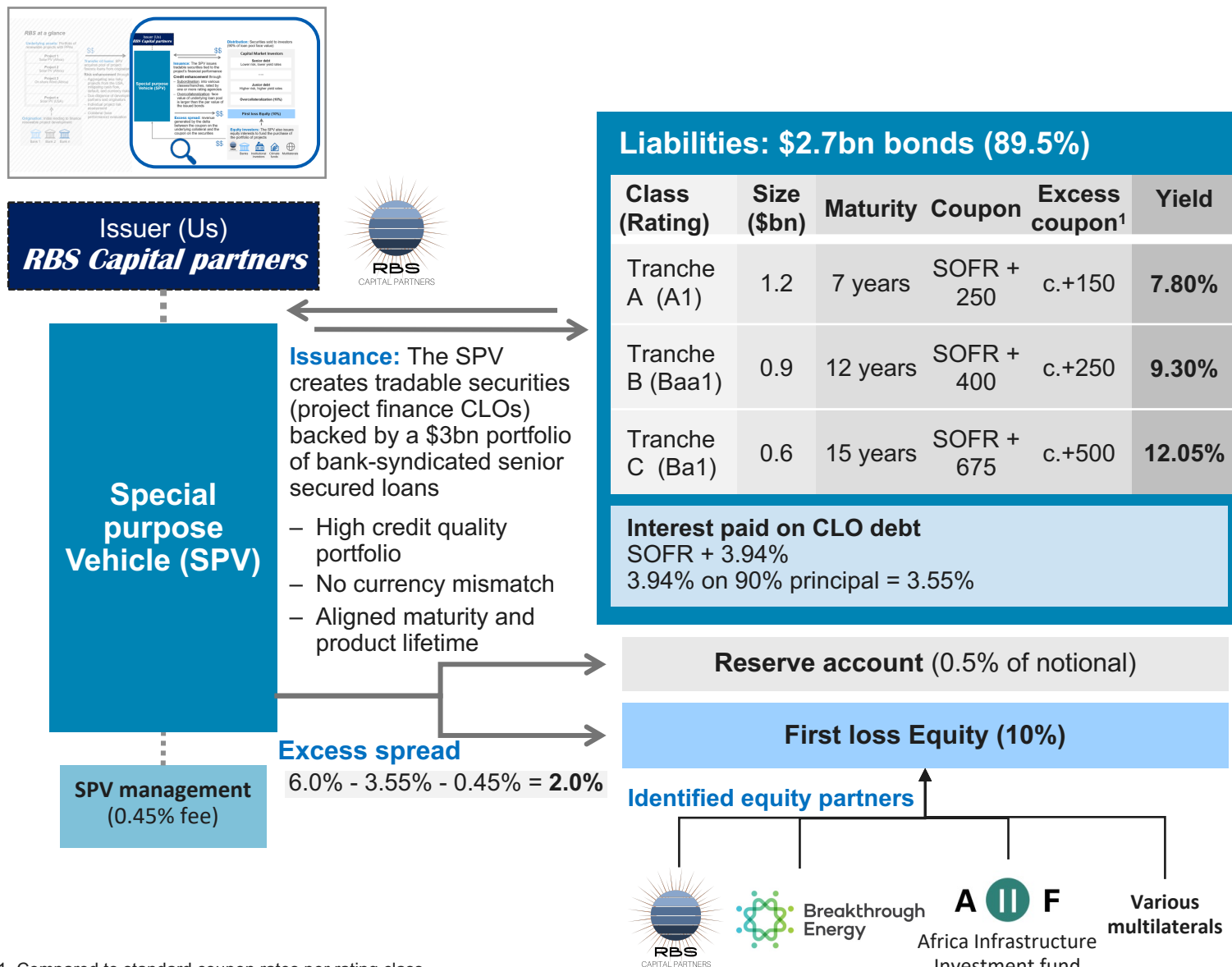
1. Oversimplification of weighted average interest rates, unifying around SOFR benchmark

Example 20 project portfolio of RBS Capital

#	Project Name	Techno-logy	Country	Capacit (MW)	Project Cost (M\$)	Debt size (M\$)	Start of operations	Lender	Developer
1	Mogalakwena	Solar PV	South Africa	100	120	84	2024	TBD	EDF / Pele Green Energy
2	Kagnas	Wind	South Africa	140	291	233	2020	ABSA	Lekela Power
3	SOLA Wheeling	Solar PV	South Africa	200	235	180	2024	Standard Bank, DBSA, Nedbank	SOLA
4	Bokamoso	Solar PV	South Africa	78	125	87	2020	Nedbank	Sun Edison
5	Garob	Wind	South Africa	145	226	181	2022	ABSA / Nedbank	Enel
6	Noor Midelt 2	Solar PV	Morocco	210	490	343	2024	TBD	Abu Dhabi Future Energy / EDF
7	Noor PV 1	Solar PV	Morocco	177	220	175	2018	Masen	Astronergy / ACWA
8	Taiba N'Diaye	Wind	Senegal	159	377	302	2021	OPIC / US Africa CEFF	Lekela
9	Kipeto	Wind	Kenya	100	320	233	2021	OPIC	BTE Renewables of SA
10	Garissa	Solar PV	Kenya	55	136	136	2019	Exim Bank of China	Kenya Rural Electrification Authority
11	Bangweulu	Solar PV	Zambia	54	60	39	2019	IFC / OPIC	Neoen / IDC Zambia / First Solar
12	Calenga	Wind	Angola	84	146	117	2025	TBD	China Energy Investment Group
13	Baia Farta	Solar PV	Angola	97	152	106	2022	Exim Bank	M. Couto Alves / Sun Africa
14	Maplewood	Solar PV	USA (Texas)	327	457	282	2021	Nord/LB / EDC / Rabobank / NBC	Canadian Solar
15	Henrietta	Solar PV	USA (California)	104	210	137	2016	Credit Agricole / Mizuho / Santander	Southern Power / Capital Dynamics
16	Ash Creek	Solar PV	USA (Texas)	408	612	588	2024	MUFG / SMBC	Primenergy
17	Sun Mountain	Solar PV	USA (Colorado)	293	352	236	2022	ING Capital / HSBC Bank USA	Lightsource BP
18	Rim Rock	Wind	USA (Montana)	189	400	320	2012	Morgan Stanley	NaturEner
19	Starr Solar	Solar PV	USA (Texas)	163	230	197	2024	ING / NatWest / Societe Generale	Lightsource BP
20	Second Div.	Solar PV	USA (Texas)	125	176	151	2024	ING / NatWest / Societe Generale	Lightsource BP



We offer attractive returns through 3 distinct tranches and first loss equity



1. Compared to standard coupon rates per rating class

Considerations

- **Tranche rating methodology:** Our fund would use existing **Moody's rating** approach for credit risk assessment for **CLOs backed by Project Finance**
- **Country risk:** simulation of country ceiling event occurrence and impact
- **Default risk:** probability of default by type of credit (e.g. PPP/PFI vs utilities); most assets in a steady-state operation can achieve high single-A ratings
- **Reserve account:** overcollateralization measure to plug in any abnormal shortfalls in cash flows with a 3-year replenishment period
- **First loss equity: Multilaterals identified as ideal partners**
 - First loss equity investors would see returns relative to their participation and would deliver a **10x impact** for their contribution
 - Traditional investors would evaluate equity as high yield debt
- **Two options for executing the loan closing and tranche issuance:**
 - **Simultaneous closing (Preferred):** we first sign with the banks we buy the loans from, then structure the tranches accordingly and make the sale to investors. RBS wouldn't require liquidity
 - **Warehousing facility (Standard):** under this mechanism, a mirror SPV would buy the loans prior to "tranching". It is very capital intensive

RBS fund offers attractive risk adjusted returns to the tranches

RBS financial assumptions

Class (Rating)	RBS Capital			Comparable market yield
	Price ¹	Coupon	Yield	
Tranche A (A1)	100	SOFR + 250	7.80%	6.22%
Tranche B (Baa1)	100	SOFR + 400	9.30%	6.77%
Tranche C (Ba1)	100	SOFR + 675	12.05%	7.04%
Equity			10.27%	12.80%

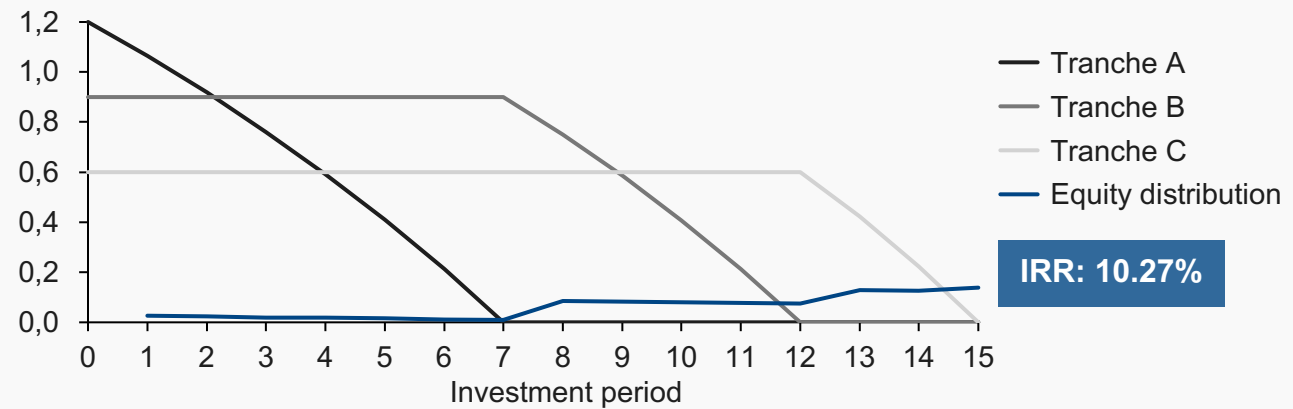
Assumptions

SOFR		SOFR (trailing 6-month avg as of close)	5.3% ²
		Fund life	15 years
Loan portfolio	Africa	African loans value (% of portfolio)	60%
		Loan interest rate	14.0%
		Cumulative default rate	7.0%
	US	Loan interest rate	SOFR + 2.0%
		Cumulative default rate	2.0%
SPV management		Annual management fees	0.45%
		Equity participation	10%
Excess Spread		Reserve account	0.5%
		Excess spread %	2.2%

1. Percentage of face value; 2. As of April 12th, 2024

RBS financial returns

Note balance and equity cashflows (\$ bn)

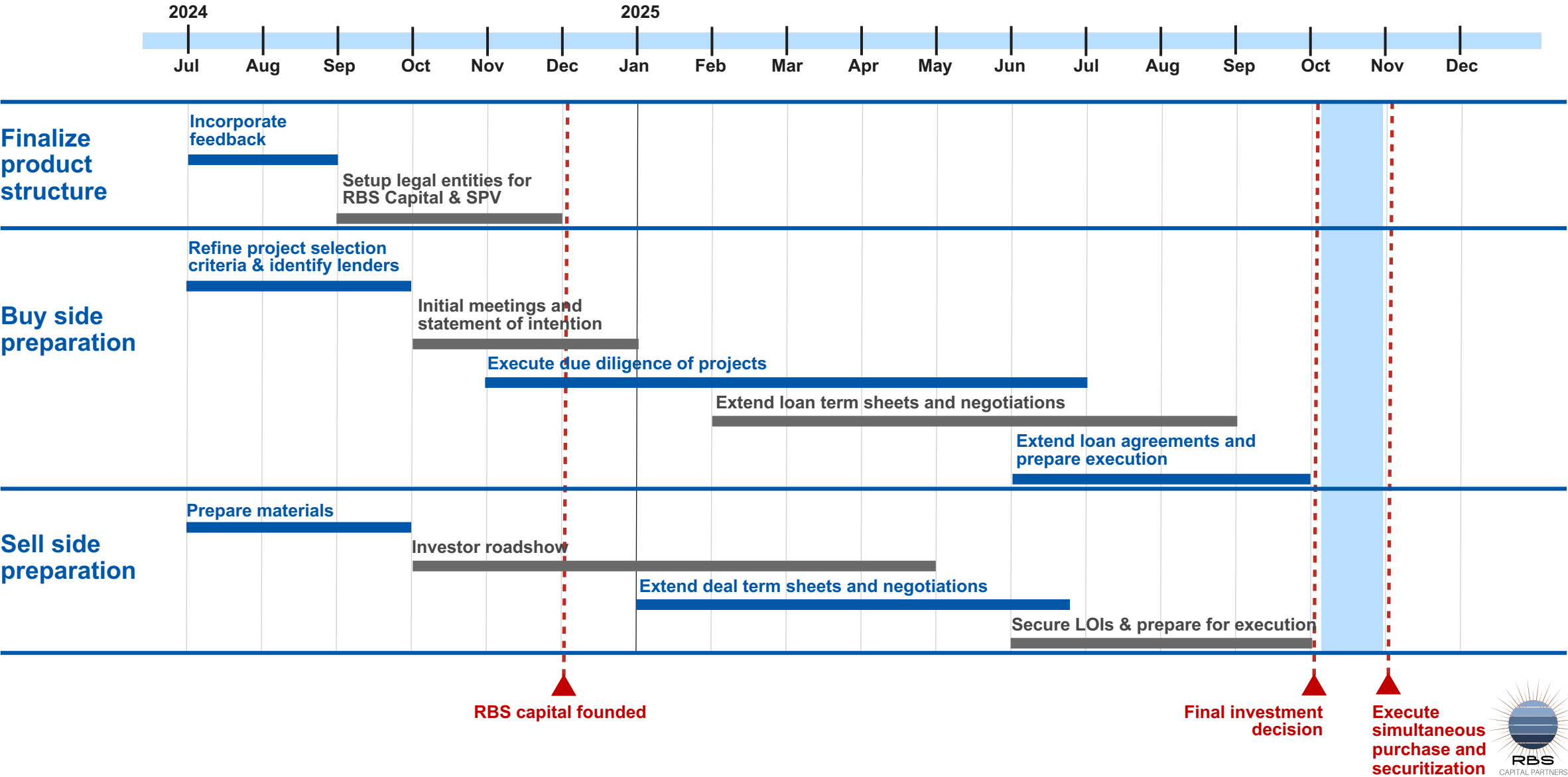


Parameter

IRR sensitivity

SOFR	60bps drop for every 25 bps SOFR increase (RBS would limit exposure to +/- 3% current SOFR)
Tranche A yield	30 bps drop for every 25 bps yield increase
Tranche B yield	50 bps drop for every 25 bps yield increase
Tranche C yield	40 bps drop for every 25 bps yield increase
Cumulative default rate on African loans	13 bps drop for every 25 bps default rate increase
Interest rate on African loans	90 bps increase every 25 bps interest rate increase

We plan to launch our product in 18 months



We foresee 4 key challenges to resolve

1 Selection of appropriate projects

 **Adverse Selection:**
We require a credible mechanism to address adverse selection. Particularly when receiving debt offers from banks

 **Collective diversity:**
Next to individual project risk mitigation, securitized risk reduction requires balanced exposure to different types of risk across the portfolio

 **Size of market:** \$3bln fund size implies we acquire ~10% of outstanding African utility-scale renewable debt. This may force us to accept sub-optimal debt in portfolio

2 Confidence building

 **Credibility as issuer:**
As first-time issuers RBS will be under increased scrutiny

 **New product category:**
Given the novelty of the product, there will be a higher cost in educating investors

3 Execution requires high level of coordination

 **Timing of execution:**
Simultaneous purchase and securitization of loans is a non-standard and precise approach which requires expert capabilities

 **Intercreditor agreements**
will be required frequently where there are multiple creditors. These can be time consuming and require high legal cost

 **Loan agreement amendments:** We expect to amend several loan agreements to achieve homogenization required in LCO

4 Impact storyline and realization

 **Second-degree impact:**
Impact is achieved through secondary market mechanisms. Achieving full impact requires repeated securitization in Africa

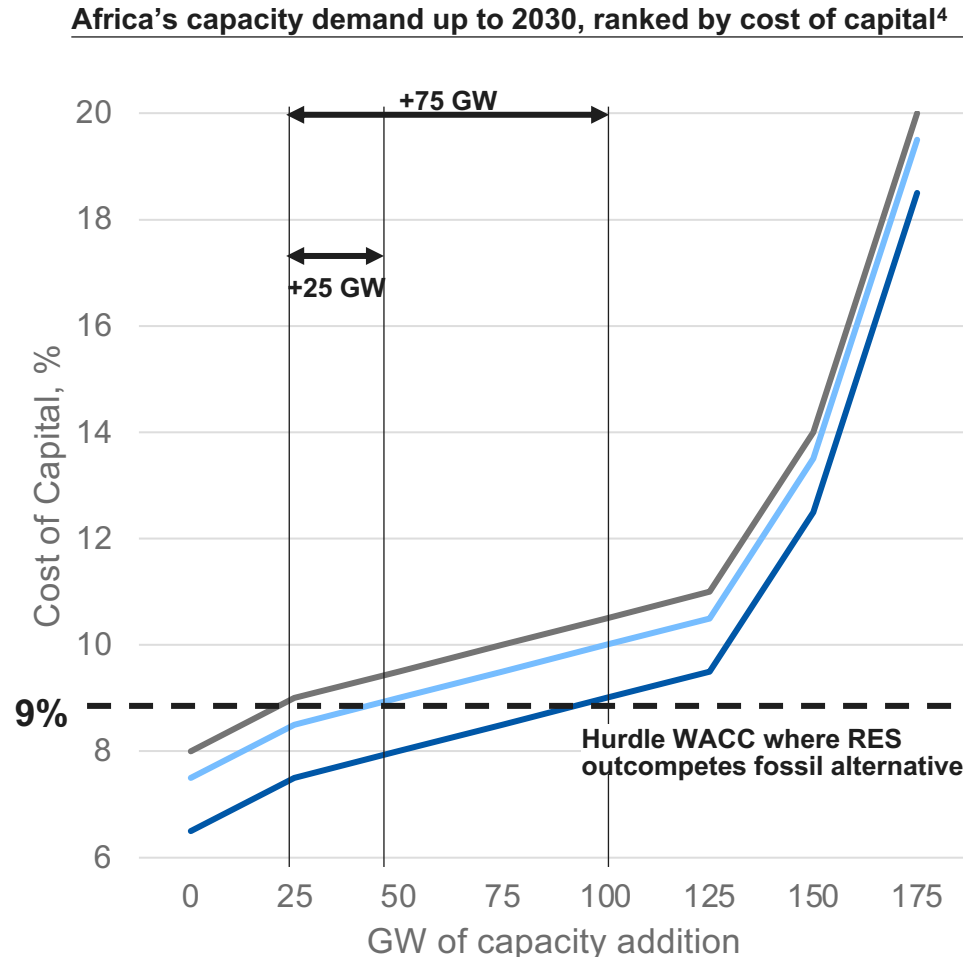
 **Appetite from first-loss equity holders:**
Our product hinges on first-loss equity holders' interest to sponsor a product that achieves only secondary impact

Our \$3bln fund size leads to 0.65-1.0 GW additional installed capacity and cost of capital effects could lead to a further 25-75 GW by 2030

1 Investment driven: Bank reinvestment of project debt

Total fund size, \$bln	3
	X
of which to African projects, %	60%
	X
of which reinvested in renewables projects, %	50-80%
	=
Additional capital invested into RES projects due to fund, \$bln	0.9-1.4
	X
Capacity realized for each invested debt dollar ¹ , MW	0.75
	=
Total additional RES capacity realized in Africa, MW	650-1040
	~
Equivalent renewable energy produced over lifetime, TWh	36-57
	X
Emission intensity gas/coal power plant, Mt/TWh ²	0.5-1.1
	=
Equivalent CO ₂ avoided, Mt	27-43 ³

2 Market driven: Improved viability of renewable projects due to lower cost of capital



Wide scale securitization is estimated to lower cost of capital of new projects by up to 1.5%⁵ with a low scenario of 0.5%

New renewable development is cost competitive at ~9% WACC

25-75 GW of cumulative capacity estimated to become competitive up to 2030 due to securitization alone

This is equivalent to 1.0-3.1 Gt of CO₂ avoided

1. Based on publicly available information of project financed RES in Africa (75% average debt) 2. International Energy Agency averages for 2022 3. Assuming a 50-50 coal/gas mix 4. IEA on total power demand up to 2030 (RES or Fossil) and cost of capital in most African countries. Curve estimated based on reported cost of capital for infrastructure projects in 2021, 2022 and 2023. Curve subject to change

Our team is excited to make RBS Capital a reality



Ignacio Aguirre

- 3 years of experience working in research and business development in Energy Industry (US Department of Energy, ComEd)
- 3+ years of experience working in Management Consulting
- MS in Mechanical Engineering and MBA Candidate at Columbia Business School

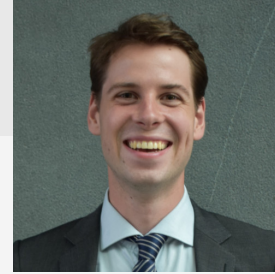
“Plans to visit Garrisa’s solar farm in Kenya this summer”



Nadim Dabbous

- 3+ years of Power and Infrastructure market experience
- MA in Economics and Infrastructure Finance and MBA Candidate at Columbia Business School

“Helped avert a black-out in New England”



Dion Koreman

- 2+ years in management consulting with a focus on renewable development & decarbonization
- 2+ years in offshore wind energy engineering
- MBA Candidate at Columbia Business School, MSc in Offshore Wind Energy, BSc in Aerospace Engineering

“Built a solar PV mini-grid in Madagascar”



Diego Rehder

- 10+ years of experience operating companies in developing markets
- Entrepreneur two-time founder
- BS in Economics and MBA Candidate at Columbia Business School

“Named his dog Watt”

Renewable Backed Securities:
**Accelerating Africa's
Energy Transition**

Thank you



Comparative LCOE analysis of a coal power plant and a solar project under different WACC rates

Project descriptions, Comparative example



Coal fired power plant

Project Lifetime: 25 years

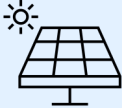
Capacity: 1000 MW

Capacity factor: 65%

CAPEX: 1000 USD/kW

Opex: 2% of CAPEX

Fuel cost: 4ct USD/kWh



Solar project

Project Lifetime: 25 years

Capacity: 100 MW

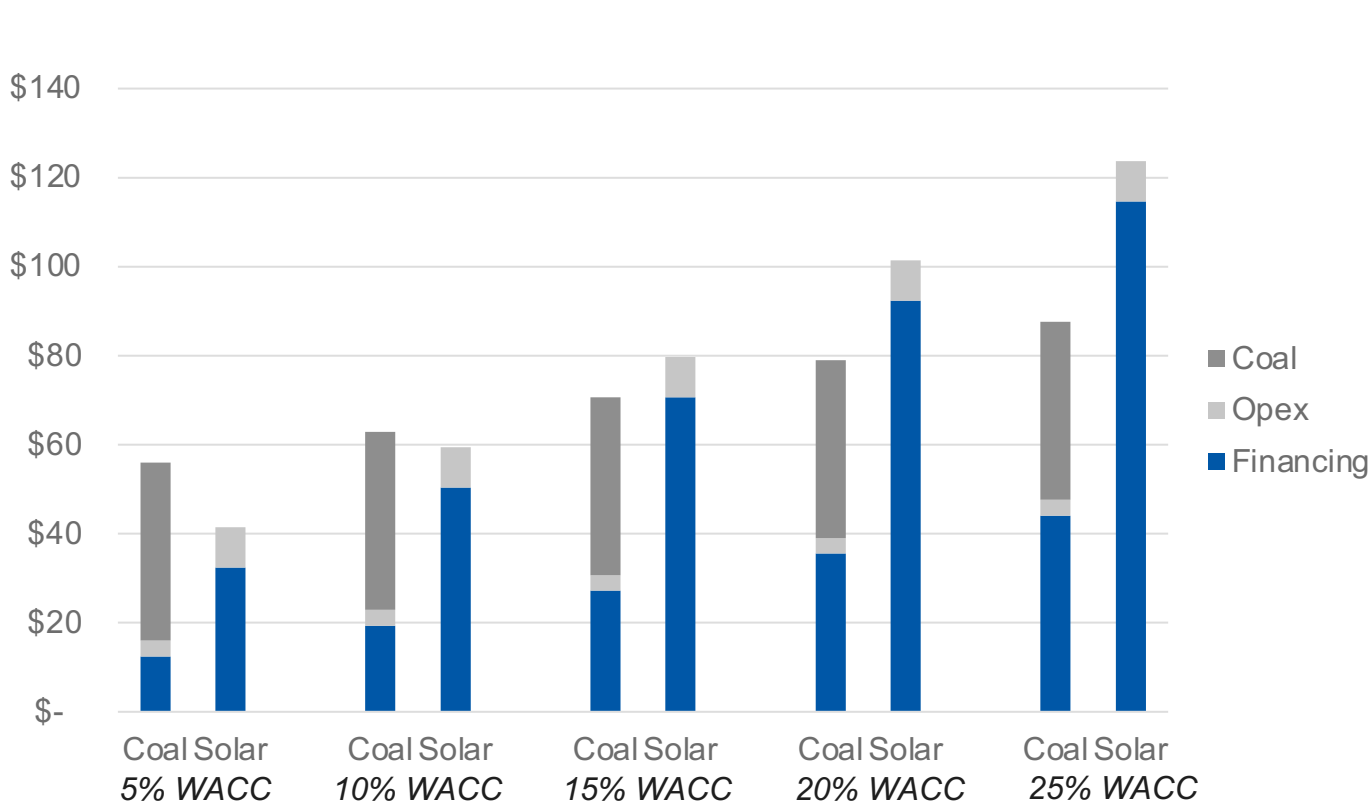
Capacity factor: 25%

CAPEX: 1000 USD/kW

Opex: 2% of CAPEX

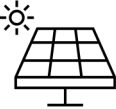
Fuel cost: -

LCOE breakdown for different cost of capital, USD/MWh



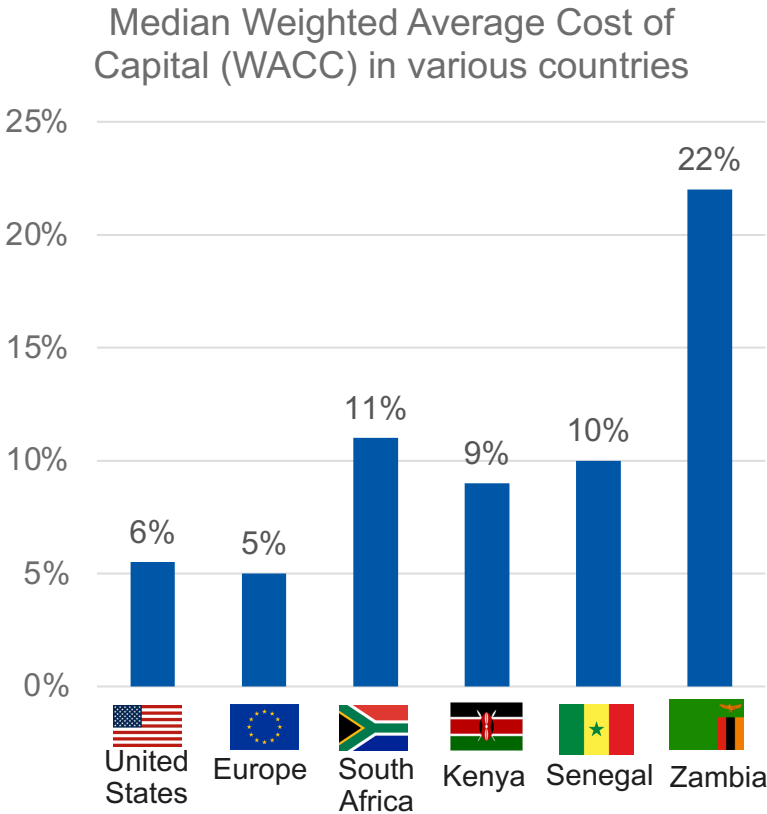
Cost of Capital determines RES competitiveness... ...and it is too high in Africa

LCOE¹ of similar coal and solar projects in the US
vs in African Countries, Comparative example

	 Unites States / Europe	 African Countries in scope
Cost of Capital ²	5-7%	9-20%
 Coal fired power plant	55-59 \$/MWh	66-88 \$/MWh
 Solar project	42-48 \$/MWh	67-124 \$/MWh

Economic advantage

Cost of Capital² guaranteed cashflow infra-projects in
selected countries, IEA 2022 & 2023, Climate Policy Initiative



Higher cost of capital renders fossil projects **more viable** in all African countries than renewable projects

In “more investible” African countries, viability **hinges on percentage points differences**



Higher cost of capital is driven by **currency, sovereign and political risk**



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1.Levelized Cost of Electricity, US\$/MWh 2. Construction phase

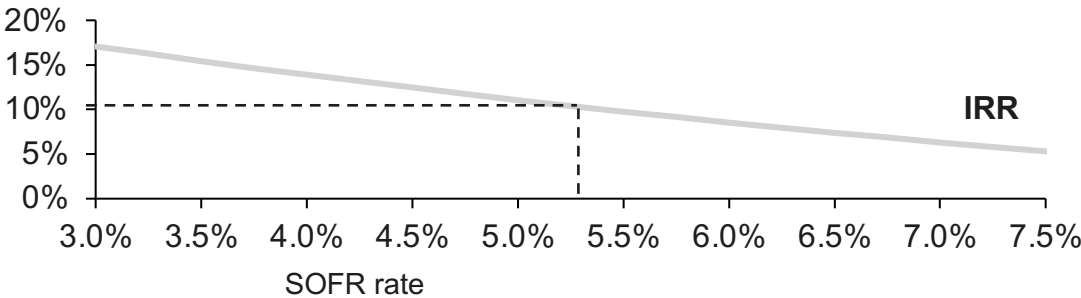
Sensitivity analysis on RBS fund returns



IRR sensitivity to Tranche A & B yields

		Tranche A yield						
		1.75%	2.00%	2.25%	2.50%	2.75%	3.00%	3.25%
Tranche B yield	3.25%	12.8%	12.4%	12.1%	11.7%	11.4%	11.0%	10.7%
	3.50%	12.3%	11.9%	11.6%	11.2%	10.9%	10.6%	10.2%
	3.75%	11.8%	11.4%	11.1%	10.7%	10.4%	10.1%	9.8%
	4.00%	11.3%	10.9%	10.6%	10.3%	9.9%	9.6%	9.3%
	4.25%	10.8%	10.4%	10.1%	9.8%	9.5%	9.2%	8.8%
	4.50%	10.3%	10.0%	9.6%	9.3%	9.0%	8.7%	8.4%
	4.75%	9.8%	9.5%	9.2%	8.9%	8.6%	8.2%	7.9%

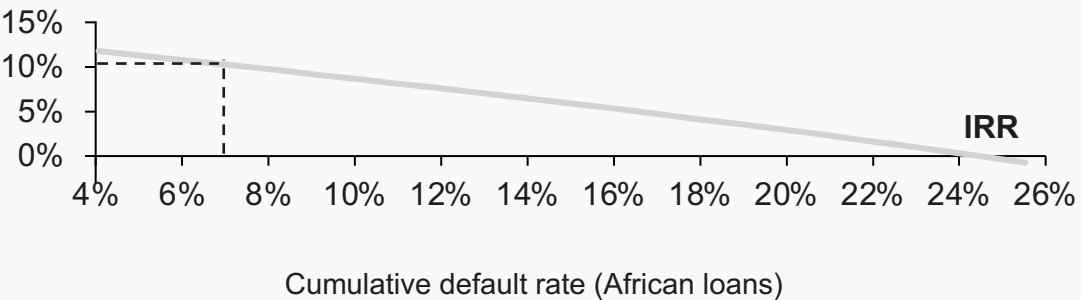
IRR vs SOFR rate



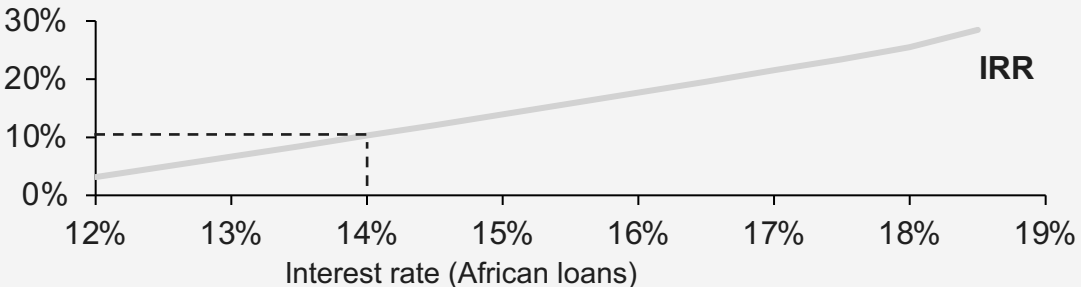
IRR sensitivity to Tranche B & C yields

		Tranche B yield						
		3.25%	3.50%	3.75%	4.00%	4.25%	4.50%	4.75%
Tranche C yield	6.00%	12.9%	12.4%	11.9%	11.4%	10.9%	10.4%	10.0%
	6.25%	12.5%	12.0%	11.5%	11.0%	10.5%	10.1%	9.6%
	6.50%	12.1%	11.6%	11.1%	10.6%	10.2%	9.7%	9.2%
	6.75%	11.7%	11.2%	10.7%	10.3%	9.8%	9.3%	8.9%
	7.00%	11.3%	10.8%	10.4%	9.9%	9.4%	9.0%	8.5%
	7.25%	10.9%	10.5%	10.0%	9.5%	9.0%	8.6%	8.1%
	7.50%	10.6%	10.1%	9.6%	9.1%	8.7%	8.2%	7.7%

IRR vs Cumulative default rate



IRR vs Interest rate on African loans



Selected Benchmark Returns

	Benchmark	Yield
First Tranche Selected Benchmarks	S&P Global Developed Bond Corporate Infrastructure Index	5.07%
	Down Jones Brookfield Global Infrastructure Broad Market Corporate Bond Index	5.3%
	Comparable energy CLO (AA1)	SOFR+2.25%
Second Tranche Selected Benchmarks	Freddie Mac Multifamily A2 Bonds	T+0.5%
	ICE BoFA AA Corporate Index AA	5.25%
	Comparable energy CLO (A3)	SOFR+4.9%
Third Tranche Selected Benchmarks	Fidelity Capital and Income	9.23%
	ICE BoFA US High Yield Index	7.87%
	Comparable Project Finance Bond Issuance (B3; CA-CEB Argentina)	10%
Equity Selected Benchmarks	VanEck Green Infrastructure ETF	-9.46%
	iShares Infrastructure ETF	5.95%
	JPM Sustainable Infra ETF	-6.69%

Our effort had contribution from renowned experts and utilized top industry sources for data inputs



- Syndication Professional



- Infrastructure Investor



- Project Finance Expert



- Lender



- Securitization Lawyer



- Debt Markets Specialist
- Currency Hedging Specialist
- Climate Finance Researcher
- Fixed Income Trader

Research

Moody's Examining Infrastructure as an Asset Class

Moody's Project Finance and Infrastructure Asset CLOs Methodology

Moody's default and recovery rates on project finance bank loans

Global Infrastructure Hub 2023 Infrastructure Monitor

S&P Annual PF Default and Recovery Study

Project Finance International 2024 Annual Report

IEA 2022 Africa Energy Outlook

TXF Intelligence Research

Guggenheim ABC's of ABS whitepaper

CA-CIB 15 project bonds to change your preconceptions

Various Fixed Income benchmarks